

10. Pattern Printing Programs

(a) Numeric Triangle Pattern

```
#include <stdio.h>
int main() {
    int i, j, num = 1;
    for(i = 1; i <= 4; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d ", num);
            num++;
        }
        printf("\n");
    }
    return 0;
}
```

Output:

```
1
2 3
4 5 6
7 8 9 10
```

(b) Character Triangle Pattern - "INFINITY"

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[] = "INFINITY";
    int len = strlen(str);
    for(int i = 0; i < len; i++) {
        for(int j = 0; j <= i; j++) {
            printf("%c ", str[j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

```
I
I N
I N F
I N F I
I N F I N
I N F I N I
I N F I N I T
I N F I N I T Y
```

(Existing sections continue below)

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